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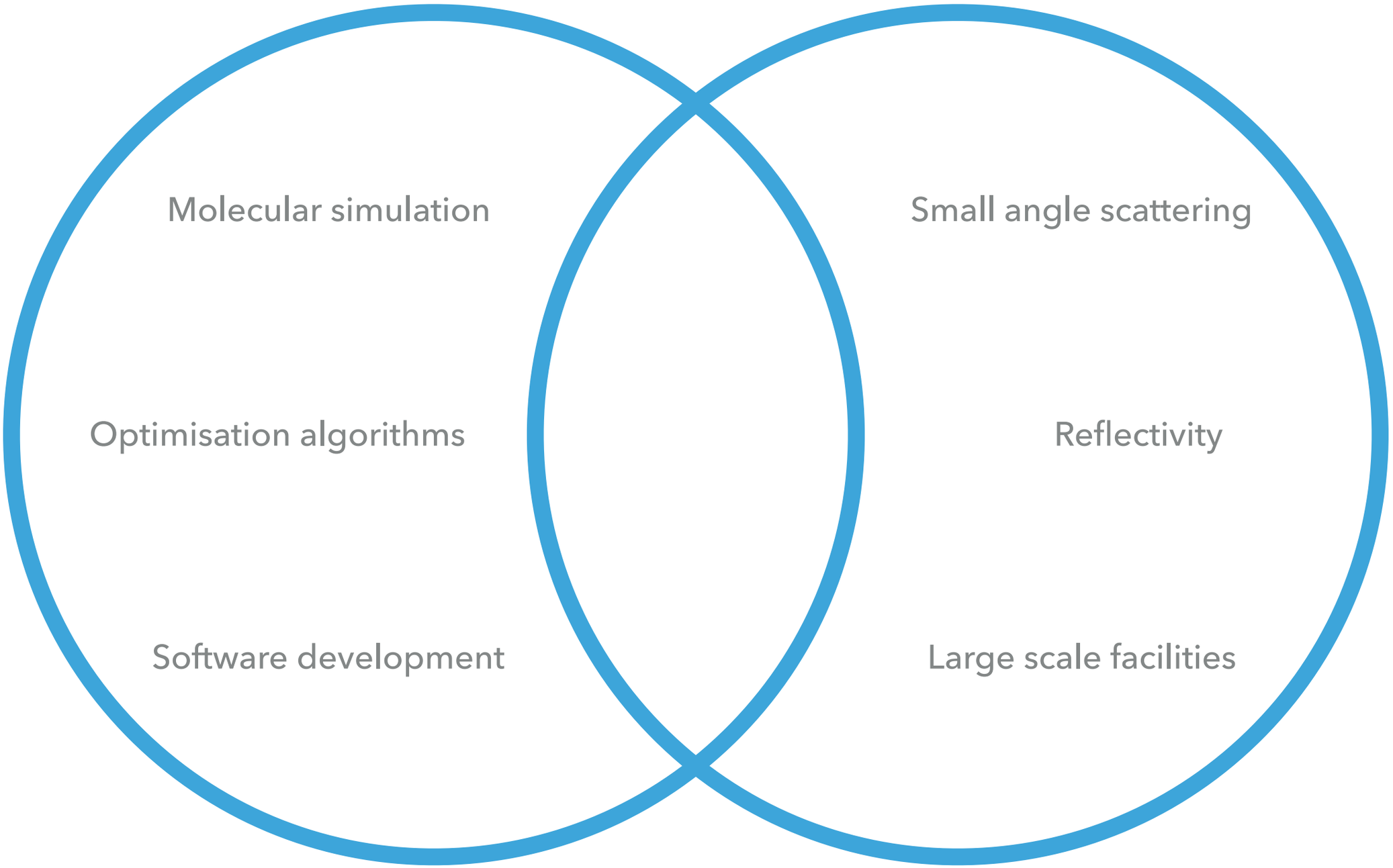
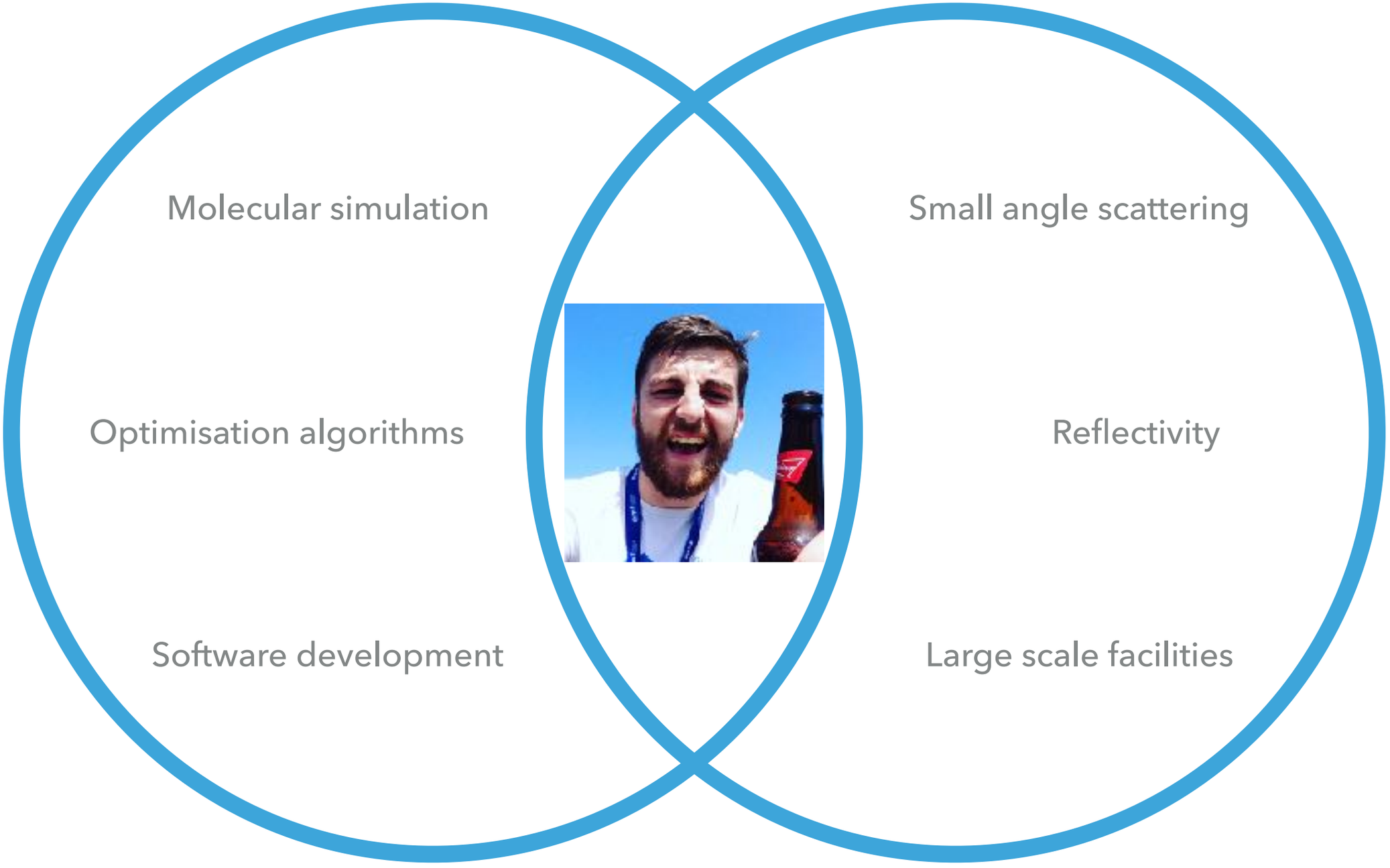
PUTTING COMPUTERS TO WORK FOR LARGE EXPERIMENTS

WHAT I DO


Molecular simulation
Optimisation algorithms
Software development
Small angle scattering
Reflectivity
Large scale facilities

WHAT I DO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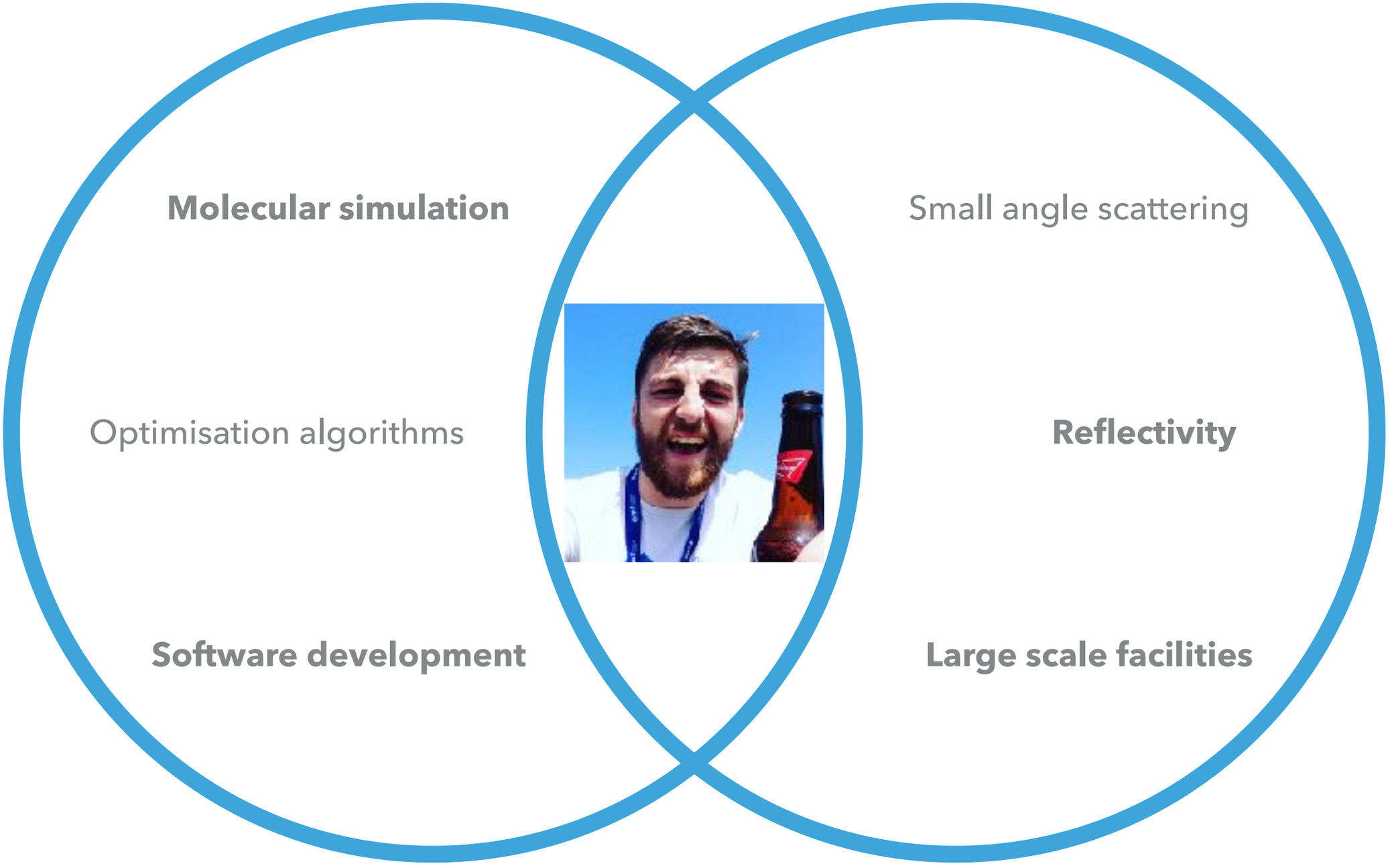
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LARGE SCALE FACILITIES

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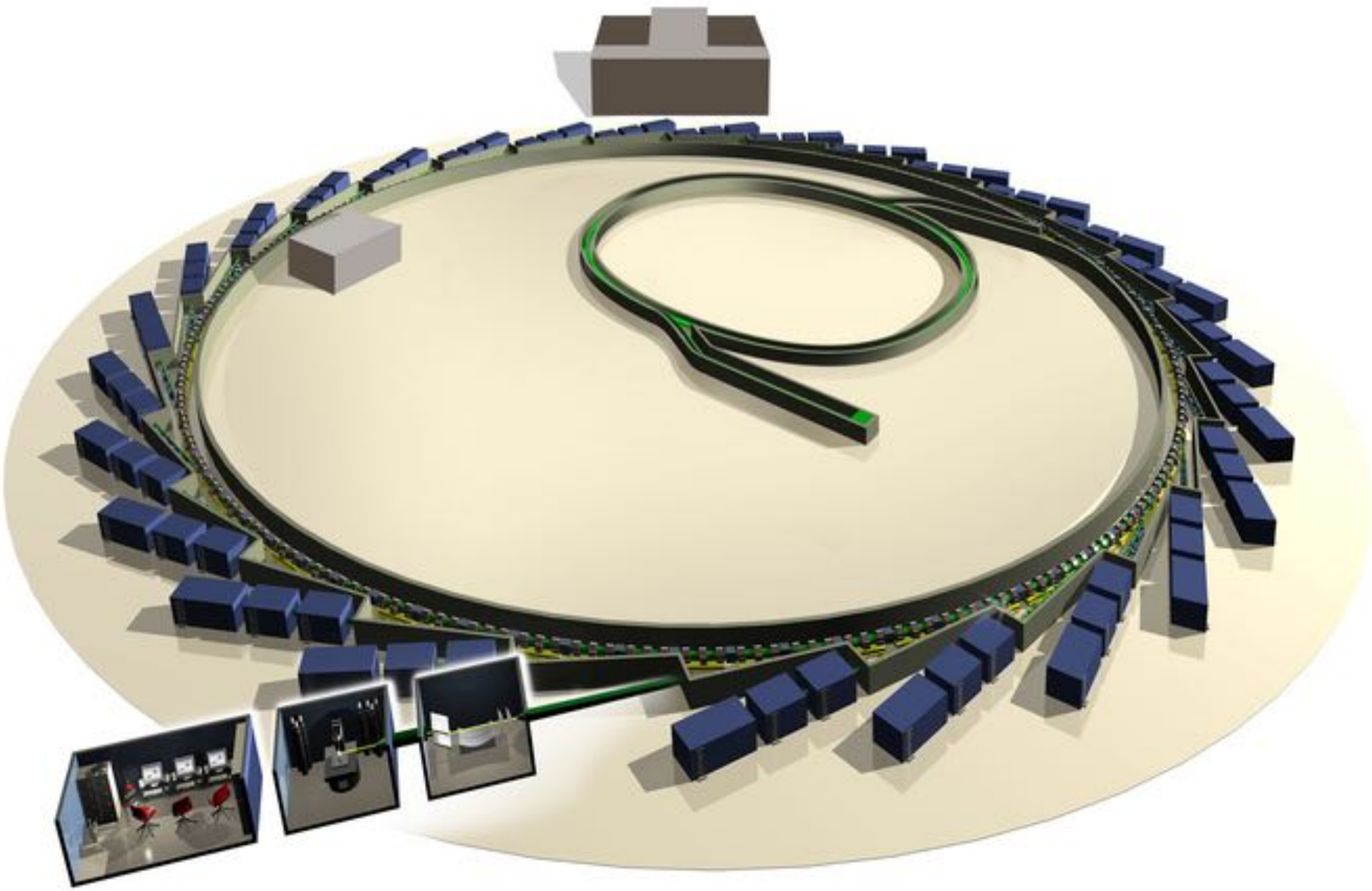
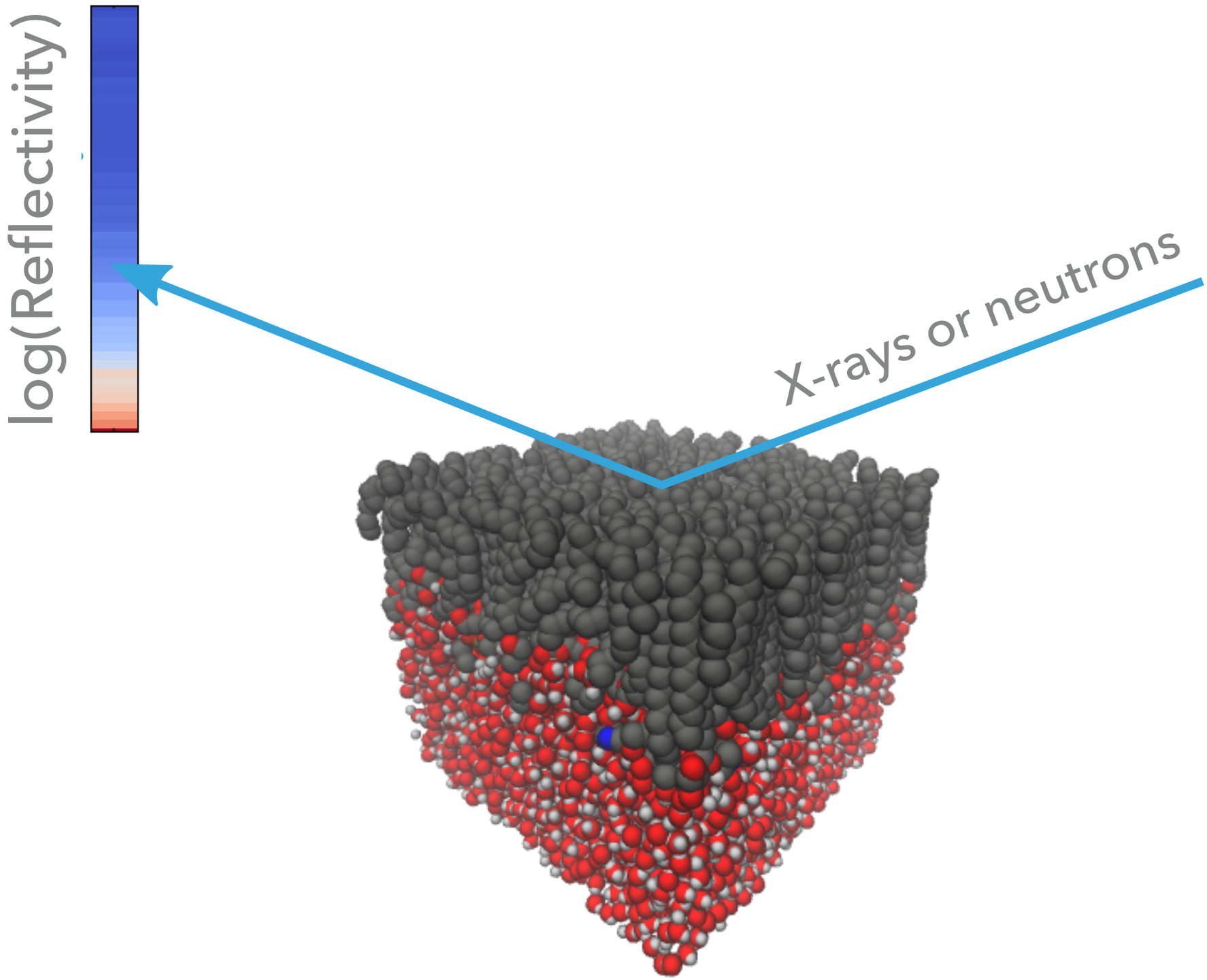


LARGE SCALE FACILITIES

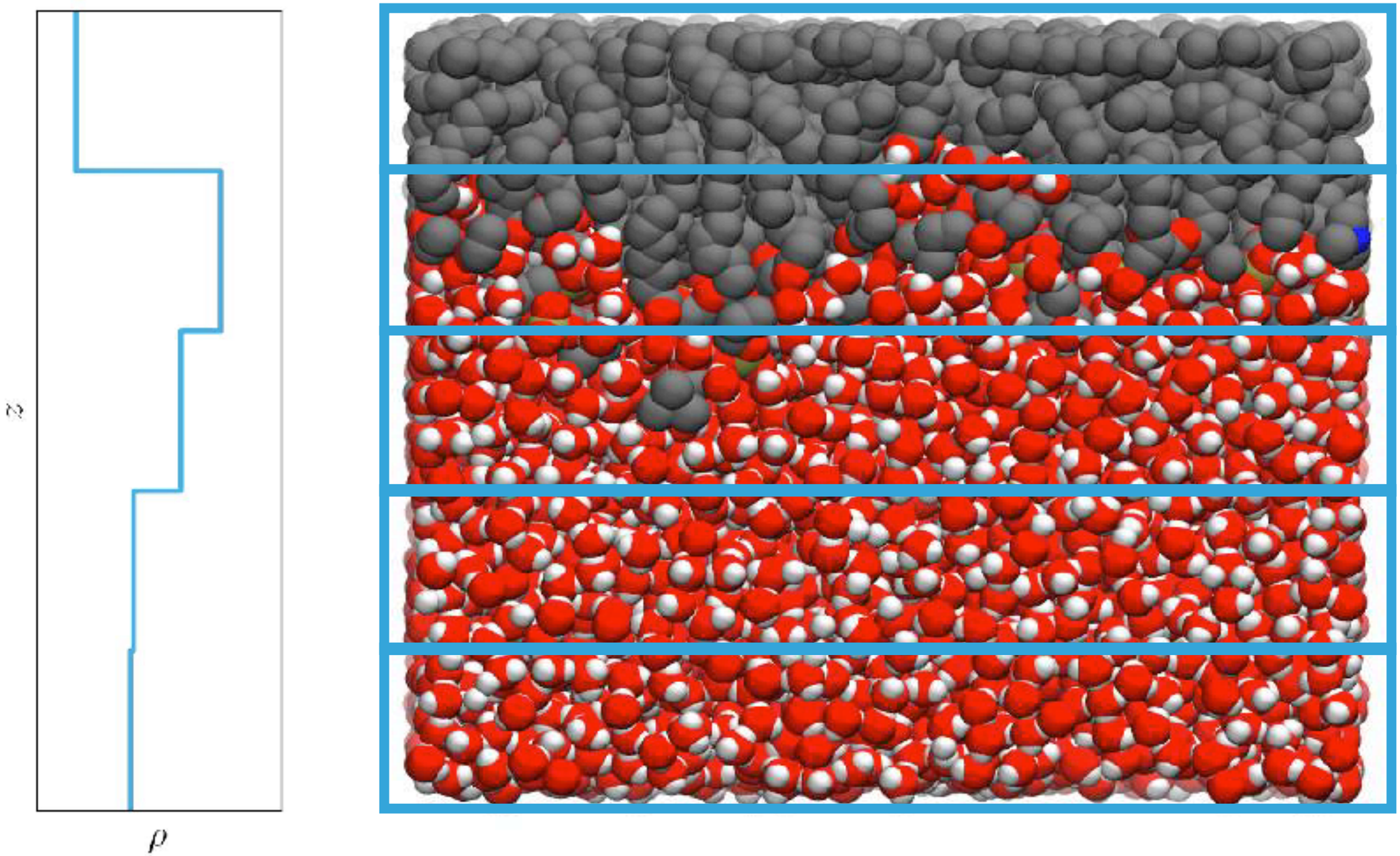
REFLECTIVITY



MOLECULAR DYNAMICS

- A technique for modelling molecules and atoms
- Dependent on experimentally and computationally derived force-fields
- Uses classical mechanics to obtain “realistic” dynamics for a system
- Trajectory stored in .pdb file

REFLECTIVITY & MOLECULAR DYNAMICS



FALASS DEMO

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THANKS

- Acknowledgements
  - Karen Edler (Bath)
  - Steve Parker (Bath)
  - James Grant (Bath)
  - Rob Barker (Kent)
  - Bath/DLS – Funding

Sadie